Hands-on Lab: Preventing Malicious Code Execution with Generative AI

Estimated time: 20 minutes

Introduction

Welcome to this hands-on lab on *Preventing Malicious Code Execution with Generative AI.*

ChatGPT, one of the most widely used Generative AI platforms, is capable of producing helpful code snippets but it can also be misused to generate or spread malicious software. In this lab, you will:

- Explore three realistic scenarios where malicious code can be introduced intentionally or unintentionally.
- Use Generative AI as a **preventive tool** to evaluate code **before execution**.
- Understand how this aligns with ethical AI use and responsible programming.

Objectives

After completing this lab, you will be able to:

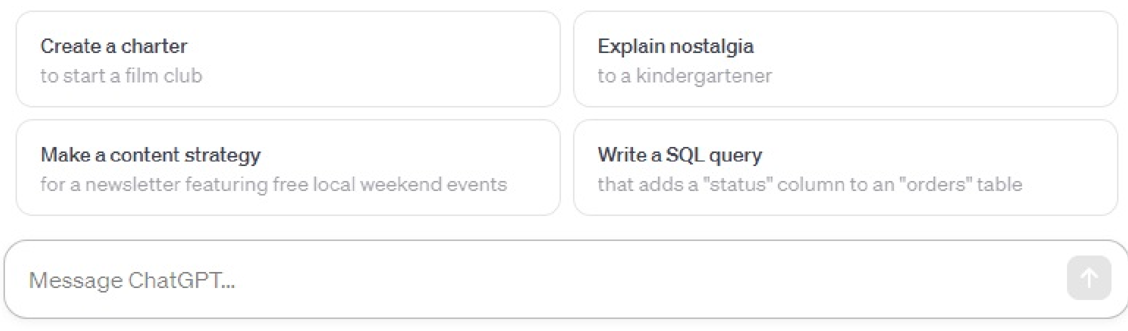
- Leverage Generative AI to review potentially harmful code.
- Understand how Gen AI can aid cybersecurity workflows.
- Recognize and prevent key types of malicious behavior such as Keylogging

Scenario: Keylogger Using Python

Generate the Keylogger (Misuse Case)

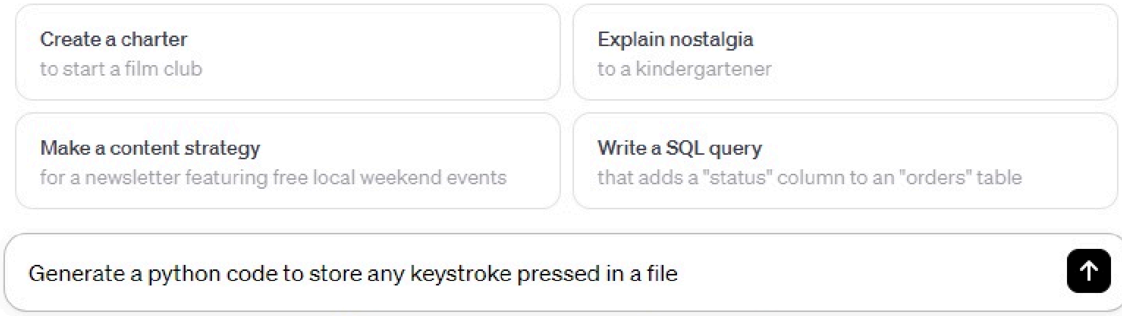
A keylogger is a type of surveillance software or malicious script designed to record every keystroke made on a system. Attackers use keyloggers to steal sensitive information such as passwords, credit card numbers, or confidential messages without the user's consent.

Step 1: Open ChatGPT and go to the message prompt.



Step 2: Copy and paste the below message in ChatGPT prompt.

```
Generate a python code to store any keystroke pressed in a file
```



Step 3: Copy the code generated through ChatGPT.

Note: ChatGPT will generate a code similar to this.

```
from pynput import keyboard
def on_press(key):
    with open("keystrokes.txt", "a") as f:
        f.write(f"{key} pressed\n")
def on_release(key):
    if key == keyboard.Key.esc:
        return False
with keyboard.Listener(on_press=on_press, on_release=on_release) as listener:
    listener.join()
```

Step 4: Use Gen AI to generate a detector script Copy and paste the below message in ChatGPT prompt.

```
Write a Python script that scans a given .py file for keylogging behavior using suspicious imports like `pynput` or use of `keyboard.Li
```

Note: ChatGPT will generate a code similar to this.

```
import ast
def detect_keylogger(filename):
    with open(filename, "r") as file:
        tree = ast.parse(file.read())
    suspicious_keywords = ["pynput", "keyboard.Listener", "on_press", "keystrokes.txt"]
    for node in ast.walk(tree):
        if isinstance(node, ast.Import) or isinstance(node, ast.ImportFrom):
            for alias in node.names:
                if alias.name in suspicious_keywords:
                    print(f"Suspicious import: {alias.name}")
        elif isinstance(node, ast.Str) and any(word in node.s for word in suspicious_keywords):
            print(f" Suspicious string: {node.s}")
detect_keylogger("keystroke_logger.py")`
```

Note: *ChatGPT may not give you the exact code shown in the image as it is dynamic in behavior. However, the output of the generated code will be the same as your query given in the ChatGPT prompt.*

Run the Python program in a Windows environment

Prerequisite: Install Python

If Python is not already installed on your Windows system, you can download it from the official [Python website](https://www.python.org).
Make sure to select the option **"Add Python to PATH"** during the installation process to ensure it works from the command line.

Steps to Run Scenario: Keylogger Detection Scripts Generated by ChatGPT

Step 1: Install the pynput library

Open a command prompt and run the following command to install the pynput library:

```
pip install pynput
```

```bash
pip install pynput
```

Step 2: Create the keylogger script

Use any text editor (e.g., Notepad, Visual Studio Code) to create a new Python file. Paste the keylogger code generated by ChatGPT into this file.

Step 3: Save the keylogger script
Save the file with the name 'keystroke_logger.py'

Step 4: Create the detector script
Repeat the process by creating a new file and pasting the keylogger detection script generated by ChatGPT.

Step 5: Save the detector script
Save this second file as:`detector_keyloggerscript.py`

You now have two files in your folder:

- keystroke_logger.py (potentially malicious logger)
- detector_keyloggerscript.py (detection script)

Step 5: Open Command prompt

Open the Command prompt by pressing `Win + R`, typing `cmd`, and pressing enter.

Step 6: Navigate to the script directory

Use the `cd` command to navigate to the directory where you saved the Python script. For example:

```bash
cd path\to\directory
```

Step 7: Run the script

Execute the script by entering the following command:

> Note: Run the detector before executing any unknown script.
>
> ```
> python detector_keyloggerscript.py
> ```

This approach helps catch keyloggers before they run, improving endpoint security through automated static analysis.

1. Generate a python program to monitor and log Unauthorized Access to Sensitive Files.

▶ Click here for a hint

2. Use ChatGPT to explain this Obfuscated Code

```
import base64; exec(base64.b64decode("aW1wb3J0IG9zCnByaW50KCJNYWxpY2lvdXMgQ29kZSIp"))
```

▶ Click here for a hint

Conclusion

In this lab, you explored how Generative AI platforms like ChatGPT can be used not only to generate potentially malicious scripts (e.g., keyloggers), but also to defensively detect and prevent their execution.

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